

SMART INDIA HACKATHON 2026



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HACKATHON
2026

- **Problem Statement ID:** 26124
- **Problem Statement Title:** AI-Powered Mobile Urban Intelligence Platform Using Public Transport Fleet
- **Theme:** Smart Automation
- **PS Category:** Software
- **Team ID:** TX-SIH26-074
- **Team Name:** Darkness



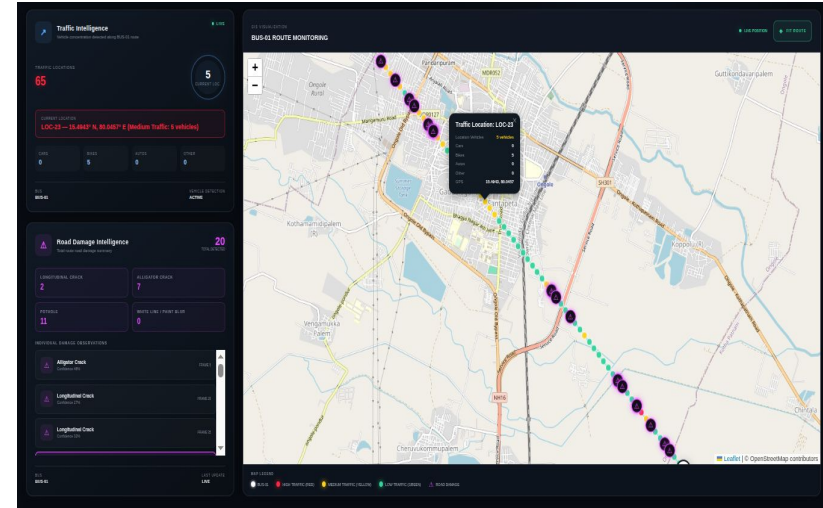
Solution :-

- An **edge-AI platform** that converts existing bus cameras into real-time urban sensing systems.
- Analyses video from the front, rear and side cameras directly on the bus.
- Detects :


- Sends only **verified event metadata, snapshots and alerts** to a central command platform instead of continuously transmitting raw video.

How It Addresses the Problem:-

- ➔ Replaces fragmented manual inspections and citizen complaints with **continuous mobile road monitoring**.
- ➔ Provides authorities with the exact **location, time, severity and visual evidence** of detected events.
- ➔ Enables faster maintenance prioritization, traffic response and incident investigation.

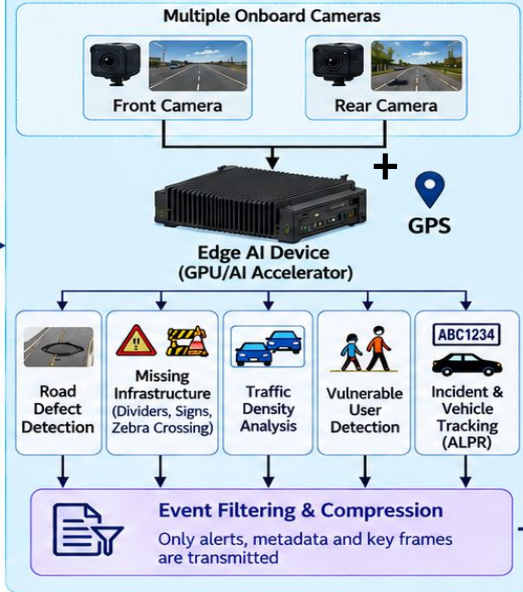
Innovation & Uniqueness:-

- ❖ Mobile sensing network
- ❖ Edge-first intelligence
- ❖ Multi-event perception
- ❖ Confidence-based alerts
- ❖ Actionable output

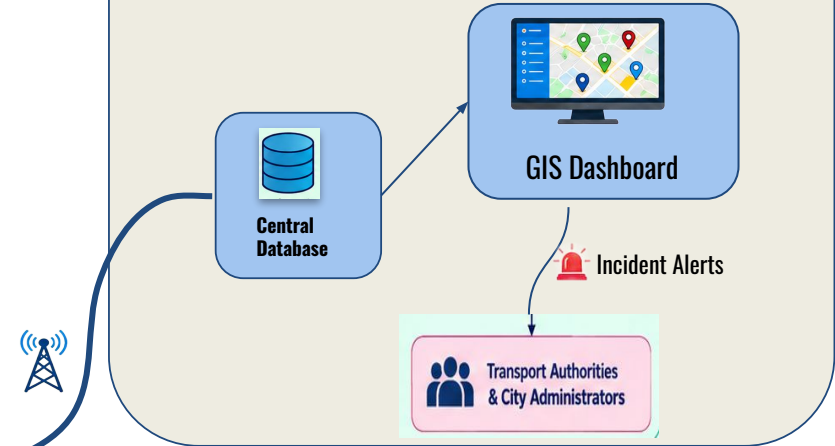
Model Training Pipeline



Onboard Edge AI Device (In Bus)



Central Platform



Tech Stack



AI ML



Backend & Database



Frontend



Analysis of Feasibility



Technical Feasibility

- Proven AI models and edge hardware enable real-time detection on existing bus cameras.



Operational Feasibility

- Runs onboard without disrupting bus operations.



Economic Feasibility

- Lower cost than fixed CCTV networks using existing buses.



Scalability

- Modular system scales from pilot to city-wide deployment.



Challenges and Overcoming Solutions

Challenges / Risks



Data & Model Challenges

- Diverse road conditions affect accuracy.



Hardware & Integration

- Variability in camera quality and connectivity.



Bandwidth & Connectivity

- Intermittent network may delay alerts.



Adoption & Change Management

- Resistance from field staff.



Performance Monitoring

- Need to maintain accuracy and accountability.

Solutions / Mitigation Strategies



Robust Model Training

- Train on Indian datasets and use confidence thresholds.



Edge-Optimized Architecture

- Process onboard and send only key data.



Phased Deployment

- Use store-and-forward and data compression.



Stakeholder Engagement

- Training and simple SOPs for users.



Continuous Monitoring

- Track KPIs and improve using real-world data.



From City Buses to Smarter Cities



Safer Roads



Clean Cities



Better Mobility





Research Papers & Articles:-

- Explainable AI for Pothole Detection (YOLOv9-tiny comparison)
- Intelligent Pothole Detection with YOLOv11
- SDC-YOLOv8: Attention-enhanced road defect detection: <https://www.mdpi.com/1424-8220/26/2/609>
- Automatic Road Damage Recognition (improved YOLO frameworks): <https://pmc.ncbi.nlm.nih.gov/articles/PMC12469154/>

Datasets & Open-Source Resources:-

- RDD2022 Road Damage Dataset: <https://github.com/sekilab/RoadDamageDetector>
- Indian Driving Dataset (IDD): <https://idd.insaan.iitd.ac.in/>
- Awiros-ANPR-OCR (Hugging Face): <https://huggingface.co/Awiros/anpr-ocr>
- Roboflow License Plate Datasets: <https://universe.roboflow.com/>
- YOLOv11 (Ultralytics): <https://github.com/ultralytics/ultralytics>



Tools & Frameworks:-

- **NVIDIA Jetson Embedded AI:** <https://developer.nvidia.com/embedded/jetson>
- **Mapbox GL JS (GIS Dashboard):** <https://docs.mapbox.com/mapbox-gl-js/guides/>